

Smart Grid System — Project Brief

Home microgrid with SMPS-based modules, droop control, and economic dispatch

Overview

- Built a home microgrid on a shared 10 V DC bus integrating PV, supercapacitor storage, grid import/export, and an LED load with a team of 7.
- Implemented module control (MPPT, commanded current, constant voltage/current) and a backend dispatch that optimizes import/export cost.
- Designed the SMPS current controller for the LED module to regulate delivered power with tight tracking accuracy.

System Architecture

Module	External Component	Topology	Control	Role
Grid Import SMPS	12 V PSU	Buck	CV, droop at 9.9 V	Sources when bus < 9.9 V
Grid Export SMPS	Sink resistor	Buck	CV, droop at 10.1 V	Sinks when bus > 10.1 V
PV SMPS	PV panel	Boost	MPPT	Harvests panel power
Supercapacitor SMPS	10.5 V to 17.5 V supercap	Bidirectional buck/boost	CC, commanded	Stores/releases on command
LED Load SMPS	3 V LED module	Buck (internal)	CC	Drives the LED load

Each droop-controlled module follows:

$$\text{Droop law: } I_{\text{module}} = \frac{V_{\text{set}} - V_{\text{bus}}}{R_{\text{droop}}}$$

with R_{droop} chosen so rated current occurs at the edge of the ± 0.1 V band, keeping regulation local and the bus stiff.

Voltage Architecture & Design Choices

A 10 V nominal bus sits above PV open-circuit minima and below supercapacitor upper bounds, while remaining reachable from a 12 V input. The 12 V PSU provides headroom without excessive switching losses typical of higher V_{in} .

$$\text{Conduction loss (matched MOSFETs): } P_{\text{cond}} = I_{\text{rms}}^2 R_{\text{ds,on}}$$

$$\text{Approx. switching loss: } P_{\text{sw}} \approx \frac{1}{2} V_{\text{in}} I_{\text{load}} (t_r + t_f) f_{\text{sw}} + \frac{1}{2} C_{\text{oss}} V_{\text{in}}^2 f_{\text{sw}}$$

$$\text{Inductor ripple (as } V_{\text{in}} \rightarrow V_{\text{bus}}): \Delta I_L \approx \frac{(V_{\text{in}} - V_{\text{bus}}) D}{L} \frac{1}{f_{\text{sw}}}$$

Dropout, not efficiency, is the binding constraint as duty approaches unity; practical limits (min off-time, gate driver refresh) prevent $D = 1$. A 12 V bench supply avoids dropout floor while staying close enough to the bus to limit switching loss.

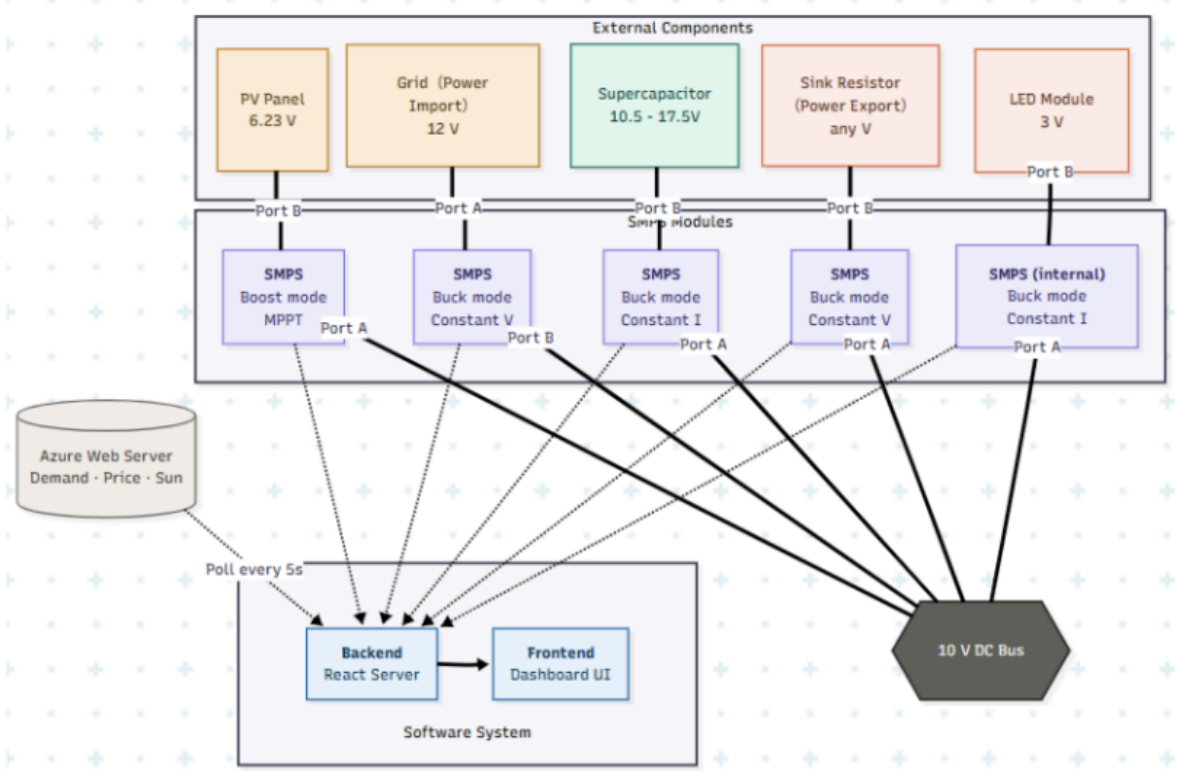


Figure 1: Overall microgrid topology: modules on a shared 10 V bus, each with local control; backend dispatch pushes setpoints.

LED Load SMPS — Design & Simulation

Three independent buck channels (red, yellow, blue), each driving a 3 W, 1 A LED, under one Pico W. Current sense uses a $0.33\ \Omega$ shunt; forward voltage is reconstructed via an external SPI ADC chain.

Control mode (Commanded Power):

$$i_{\text{ref}} = \frac{P_{\text{demand}}}{v_{\text{LED}}}$$

tracked by a per-channel PI controller, clamped to ratings (3.0 W, 1.05 A) and guarded by a forward-voltage floor when strings are dark.

Testing & Implementation

Firmware was refactored to remove external PID dependency, unify gains with bidirectional modules, and add protection (trip at $1.4\times$ rated current). Startup checks and watchdog logic improve diagnosability and safe shutdown order.

Requirements Verification

Req	Requirement	Realised by	Verification	Status
1	Supply domestic LEDs per web demand	LED Load SMPS	Power-tracking test	Implemented
2	Extract PV energy with MPPT	PV SMPS	MPPT mode comparison	Implemented
3	Store excess in supercapacitor	Supercap SMPS	Cycle test; round-trip	Implemented
4	Import/export to grid, metered	Import/Export SMPS	Efficiency sweep	Implemented
5	Minimise cost and beat naive	Algorithm	Dry run vs baseline	Implemented; 20% improvement
6	UI for current/historic energy	Web dashboard	Live plots	Implemented

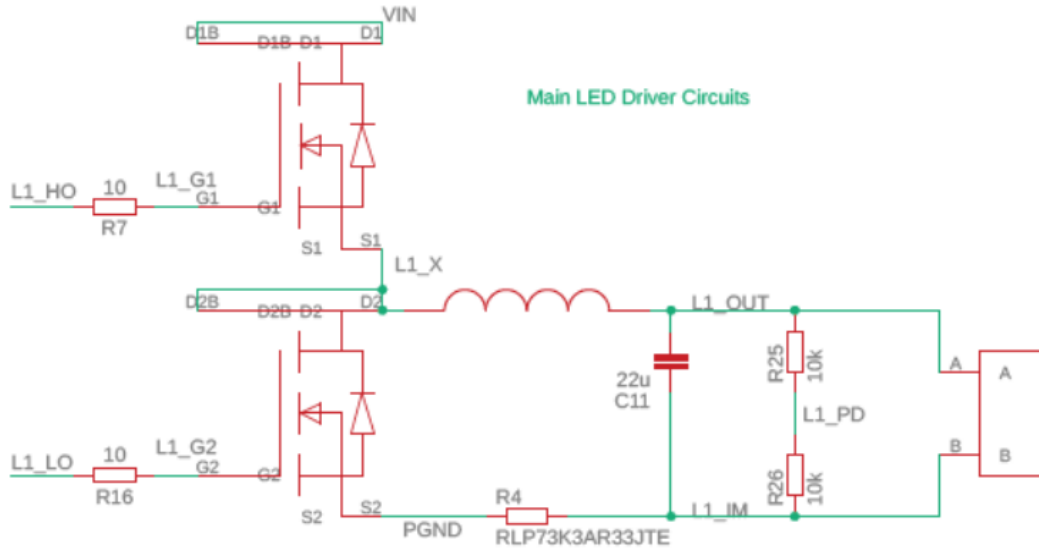


Figure 2: LED buck driver stage (one of three identical channels).

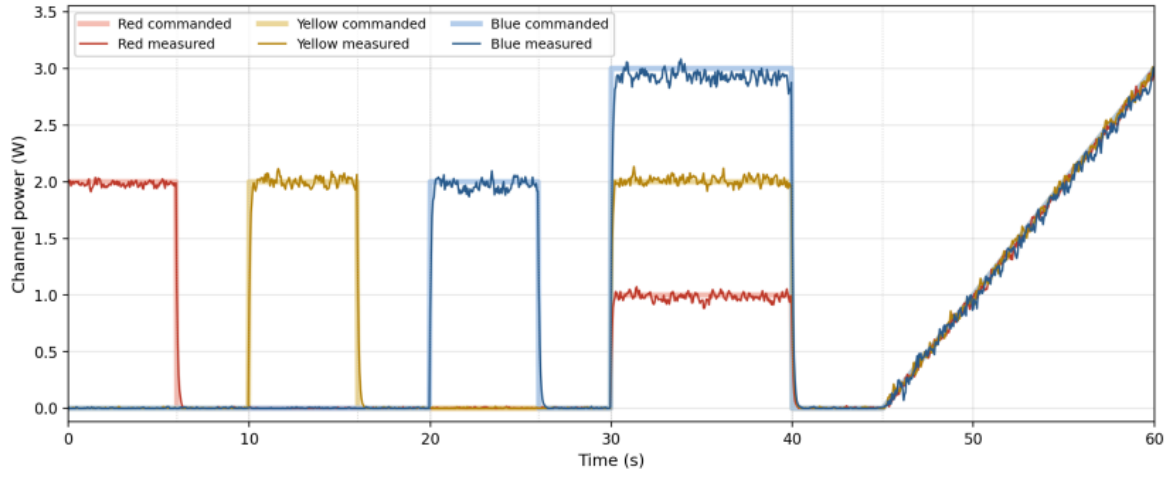


Figure 3: Commanded vs measured LED power over scripted sequences; tracking within $\pm 3\%$ across operating points.

Key outcomes:

- Stable 10 V bus with droop coordination across import/export converters.
- LED module tracks demanded power within $\pm 3\%$ across sequences and ramps.
- Economic dispatch improves cost by $\approx 20\%$ vs naive baseline.

Tools: Python/FastAPI (backend), React (dashboard), LTspice/Matlab (analysis).

Role: SMPS schematics/interconnect, LED current controller design and implementation, system integration/testing.